

Table S2

Protein	Uniprot ID	Function	Name	References
GCK	P35557	Enzyme	Glucokinase	Gersing, S., Schulze, T. K., Cagiada, M., Stein, A., Roth, F. P., Lindorff-Larsen, K., & Hartmann-Petersen, R. (2024). Characterizing glucokinase variant mechanisms using a multiplexed abundance assay. <i>Genome biology</i> , 25(1), 98.
PTEN	P60484	Enzyme	Phosphatase and tensin homolog	Matreyek, K. A., Starita, L. M., Stephany, J. J., Martin, B., Chiasson, M. A., Gray, V. E., ... & Fowler, D. M. (2018). Multiplex assessment of protein variant abundance by massively parallel sequencing. <i>Nature genetics</i> , 50(6), 874-882.
VKOR	Q9BQB6	Enzyme	Vitamin K expoxide reductase	Chiasson, M. A., Rollins, N. J., Stephany, J. J., Sitko, K. A., Matreyek, K. A., Verby, M., ... & Fowler, D. M. (2020). Multiplexed measurement of variant abundance and activity reveals VKOR topology, active site and human variant impact. <i>elife</i> , 9, e58026.
NUDT15	Q9NV35	Enzyme	Nucleotide triphosphate diphosphatase	Suiter, C. C., Moriyama, T., Matreyek, K. A., Yang, W., Scaletti, E. R., Nishii, R., ... & Yang, J. J. (2020). Massively parallel variant characterization identifies NUDT15 alleles associated with thiopurine toxicity. <i>Proceedings of the National Academy of Sciences</i> , 117(10), 5394-5401.
ASPA	P45381	Enzyme	Aspartoacylase	Grønbaek-Thygesen, M., Voutsinos, V., Johansson, K. E., Schulze, T. K., Cagiada, M., Pedersen, L., ... & Hartmann-Petersen, R. (2024). Deep mutational scanning reveals a correlation between degradation and toxicity of thousands of aspartoacylase variants. <i>Nature Communications</i> , 15(1), 4026.
TPMT	P51580	Enzyme	Thiopurine S-methyltransferase	Matreyek, K. A., Starita, L. M., Stephany, J. J., Martin, B., Chiasson, M. A., Gray, V. E., ... & Fowler, D. M. (2018). Multiplex assessment of protein variant abundance by massively parallel sequencing. <i>Nature genetics</i> , 50(6), 874-882.
CYP2C9	P11712	Enzyme	Cytochrome P450 enzyme	Amorosi, C. J., Chiasson, M. A., McDonald, M. G., Wong, L. H., Sitko, K. A., Boyle, G., ... & Dunham, M. J. (2021). Massively parallel characterization of CYP2C9 variant enzyme activity and abundance. <i>The American Journal of Human Genetics</i> , 108(9), 1735-1751.
OCT	O15245	Transporter	Organic cation transporter 1	Yee, S. W., Macdonald, C. B., Mitrovic, D., Zhou, X., Koleske, M. L., Yang, J., ... & Coyote-Maestas, W. (2024). The full spectrum of SLC22 OCT1 mutations illuminates the bridge between drug transporter biophysics and pharmacogenomics. <i>Molecular cell</i> , 84(10), 1932-1947.
PRKN	O60260	Enzyme	E3 ubiquitin-protein ligase parkin	Clausen, L., Okarmus, J., Voutsinos, V., Meyer, M., Lindorff-Larsen, K., & Hartmann-Petersen, R. (2024). PRKN-linked familial Parkinson's disease: cellular and molecular mechanisms of disease-linked variants. <i>Cellular and Molecular Life Sciences</i> , 81(1), 223.

Table S3

Protein	UniProt ID	Active Site Structure	Active Site Substrate	Active Site Annotation	Allosteric Site Structure	Allosteric Modulator	Allosteric Site Annotation	Type	Slope	Half Distance	t-value	P-value	Adj R2
GCK	P35557	1V4S, 3FGU	ANP,BGC, GLC	from 1V4S	1V4S,3ID8	MRK (activator)	literature (Kamata et al., 2004) (Larion & Miller, 2009)	kinase	-0.04	13.88	-9.01	<2e-16	0.16
PTP1B	P18031	1PTY	PTR	from 1PTY	1T48,1T49,1T4J	892,BB3,FRJ (inhibitor)	consensus	phosphatase	-0.02	13.99	-4.08	5.97E-05	0.05
PDK1	O15530	4RQK, 5LVM	ADE,ATP	consensus	3ORZ,4RQK	2A2 (activator), R1S (inhibitor)	from 4RQK	kinase	-0.05	8.89	-7.84	9.39E-14	0.18
CHK1	O14757	2E9N, 5OOP	76A,ANP	consensus	3F9N	38M (inhibitor)	from 3F9N	kinase	-0.04	11.73	-6.9	3.95E-11	0.15
CASP1	P29466	2HBQ, 6BZ9	PI (chain C)	consensus	2FQQ	F1G (inhibitor)	from 2FQQ	protease	-0.03	13.31	-4.2	3.74E-05	0.065
IDH1	O75874	5YFN, 6ADG, 6B0Z, 6BKZ, 6IOO	FLC,ICT,MG, NAP,NDP	consensus from 6B0Z, 6BKZ, 6IOO	5SUN,6B0Z, 6BKZ,6BL2,6IOO	70Q, AOU, C81, CIT, DWM, DWS (inhibitor)	consensus from 6B0Z, 6BKZ, 6IOO	oxidoreductase	-0.03	12.42	-6.39	5.32E-10	0.1

Table S4. PSD-95-PDZ3: linear regression models

	Model Comparison	Res.Df	RSS	Sum of Sq	F	Pr(>F)
whole protein	ESM1v ~ ΔΔGf	1565	21,260			
whole protein	ESM1v ~ ΔΔGf + ΔΔGb	1564	18,199	3,061.20	263.07	< 2.2e-16 ***
non-orthosteric sites	ESM1v ~ ΔΔGf	1323	15,721			
non-orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb	1322	13,441	2279.3	224.18	< 2.2e-16 ***
orthosteric sites	ESM1v ~ ΔΔGf	240	1845			
orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb	239	1704.8	140.21	19.657	1.413e-05 ***

Table S5. GRB2-SH3: linear regression model

	Model Comparison	Res.Df	RSS	Sum of Sq	F	Pr(>F)
whole protein	ESM1v ~ ΔΔGf	754	7,968			
whole protein	ESM1v ~ ΔΔGf + ΔΔGb	753	5,795	2,173.30	282.41	< 2.2e-16 ***
non-orthosteric sites	ESM1v ~ ΔΔGf	583	3,265			
non-orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb	582	3,020	245.48	47.311	1.568e-11 ***
orthosteric sites	ESM1v ~ ΔΔGf	169	1864.3			
orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb	168	1343	521.27	65.205	1.245e-13 ***

Table S6. PSD-95-PDZ3: linear mixed models

	Model Comparison	npar	AIC	BIC	logLik	-2logLik	Chisq	Df	Pr(>Chisq)
whole protein	ESM1v ~ ΔΔGf + (1 pos)	4	7575.2	7596.6	-3783.6	7567.2			
whole protein	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	7409.7	7436.5	-3699.9	7399.7	167.48	1	< 2.2e-16 ***
non-orthosteric sites	ESM1v ~ ΔΔGf + (1 pos)	4	6354.3	6375.1	-3173.2	6346.3			
non-orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	6215.8	6241.7	-3102.9	6205.8	140.57	1	< 2.2e-16 ***
orthosteric sites	ESM1v ~ ΔΔGf + (1 pos)	4	1175.3	1189.2	-583.64	1167.3			
orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	1155.6	1173	-572.78	1145.6	21.705	1	3.179e-06 ***

Table S7. GRB2-SH3: linear mixed models

	Model Comparison	npar	AIC	BIC	logLik	-2logLik	Chisq	Df	Pr(>Chisq)
whole protein	ESM1v ~ ΔΔGf + (1 pos)	4	3339.9	3358.4	-1666	3331.9			
whole protein	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	3304.1	3327.2	-1647	3294.1	37.858	1	7.609e-10 ***
non-orthosteric sites	ESM1v ~ ΔΔGf + (1 pos)	4	2461.3	2478.8	-1226.7	2453.3			
non-orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	2434.5	2456.4	-1212.3	2424.5	28.795	1	8.047e-08 ***
orthosteric sites	ESM1v ~ ΔΔGf + (1 pos)	4	810.36	822.92	-401.18	802.36			
orthosteric sites	ESM1v ~ ΔΔGf + ΔΔGb + (1 pos)	5	807.81	823.52	-398.91	797.81	4.5424	1	0.03307 *